This file contains the test of the out of the box CNN on our test set

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import os
import csv
from pathlib import Path
import torch
from PIL import Image as PILImage
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
from transformers import AutoImageProcessor, AutoModelForImageClassification
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
```

```
ROOT_DATASET = "/content/drive/MyDrive/Project 1-1/Dataset/Images/Test"
MODEL_IDENTIFIER = "Ateeqq/ai-vs-human-image-detector"
VALID_EXT = {".jpg", ".jpeg", ".png", ".bmp", ".gif", ".webp"}
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")
PARENT_TO_LABEL = {
    "ai": 1,
    "not_ai": 0,
}
```

```
Using device: cpu
```

```
processor = AutoImageProcessor.from_pretrained(MODEL_IDENTIFIER)
model = AutoModelForImageClassification.from_pretrained(MODEL_IDENTIFIER).to(device)
model.eval()
```

Show hidden output

```
def get_label_from_path(path):
    folder = Path(path).name.strip().lower()
    return PARENT_TO_LABEL.get(folder, None)
```

```
y_true = []
y_pred = []

print(f"Scanning dataset: {ROOT_DATASET}\n")

for root, dirs, files in os.walk(ROOT_DATASET):
    label = get_label_from_path(root)

    if label is None:
        continue

    for filename in sorted(files):
        image_path = os.path.join(root, filename)

        ext = os.path.splitext(filename)[1].lower()
        if ext not in VALID_EXT:
            print(f"[SKIP FILE - EXTENSION] {image_path}")
            continue

        try:
            with PILImage.open(image_path) as img:
                image = img.convert("RGB")
        except Exception as e:
            print(f"[SKIP FILE - OPEN ERROR] {image_path} | {e}")
            continue

        try:
            inputs = processor(images=image, return_tensors="pt")
            inputs = {k: v.to(device) for k, v in inputs.items()}
        except Exception as e:
```

```
            print(f"[SKIP FILE - PREPROCESS ERROR] {image_path} | {e}")
            continue

        try:
            with torch.no_grad():
                outputs = model(**inputs)
                logits = outputs.logits
        except Exception as e:
            print(f"[SKIP FILE - INFERENCE ERROR] {image_path} | {e}")
            continue


        pred_idx = int(logits.argmax(-1).item())

        id2label = getattr(model.config, "id2label", None) or {}
        pred_label_name = id2label.get(pred_idx, f"label_{pred_idx}")

        # Map prediction to binary label
        name_lc = pred_label_name.lower()
        if name_lc.startswith("hum") or "human" in name_lc or "real" in name_lc:
            pred_label = 0
        elif name_lc.startswith("ai") or "generated" in name_lc or "synthetic" in name_lc:
            pred_label = 1
        else:
            if logits.shape[-1] == 2:
                pred_label = int(pred_idx)

        y_true.append(int(label))
        y_pred.append(pred_label)
```

Show hidden output

```
print("\n=== METRICS ===")

acc = accuracy_score(y_true, y_pred)
prec = precision_score(y_true, y_pred, zero_division=0)
rec = recall_score(y_true, y_pred, zero_division=0)
f1 = f1_score(y_true, y_pred, zero_division=0)

print(f"Accuracy:  {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall:    {rec:.4f}")
print(f"F1 Score:  {f1:.4f}")

cm = confusion_matrix(y_true, y_pred, labels=[1,0])
print("\nConfusion Matrix (rows=true [ai,human], cols=pred [ai,human]):")
print(cm)
```

```
=== METRICS ===
Accuracy:  0.5000
Precision: 0.4390
Recall:    0.9000
F1 Score:  0.5902

Confusion Matrix (rows=true [ai,human], cols=pred [ai,human]):
[[18  2]
 [23  7]]
```

```
ai_count = y_true.count(1)
not_ai_count = y_true.count(0)

print(f"\nClass Balance (True Labels):")
print(f"  AI instances: {ai_count}")
print(f"  Not AI instances: {not_ai_count}")
```

```
Class Balance (True Labels):
  AI instances: 20
  Not AI instances: 30
```

```
import matplotlib.pyplot as plt
from PIL import Image

print("\n=== Displaying Example Images ===")
for i, example in enumerate(print_examples):
```

```
        image_path = example['image_path']
        true_name = 'AI' if example['true_label'] == 1 else 'Human'
        pred_name = 'AI' if example['predicted_label'] == 1 else 'Human'

        try:
            img = Image.open(image_path)

            plt.figure(figsize=(6, 6))
            plt.imshow(img)
            plt.title(f"Example {i+1}: True: {true_name}, Predicted: {pred_name}\nImage: {image_path.split('/')[-1]}")
            plt.axis('off')
            plt.show()

        except Exception as e:
            print(f"Could not display image {image_path}: {e}")
        print("----------------------------------------")
```

```
=== Displaying Example Images ===
```

Example 1: True: AI, Predicted: AI
Image: abnerfigueroa_2987323493801148116_2.jpg



Example 2: True: AI, Predicted: Human
Image: adif_robotics_3709074872061726134_10.JPG



Example 3: True: AI, Predicted: AI
Image: ai.art.sy_3708048238920477728_1.jpg





Example 4: True: AI, Predicted: AI
Image: ai.coralayne_3696337544787785468_2.WEBP



Example 5: True: AI, Predicted: AI
Image: ai.dreaming.maiden_3708959139688223050_5.jpg



Example 6: True: AI, Predicted: AI
Image: ai_arthijabi_3697411615411877513_7.jpg





Example 7: True: AI, Predicted: AI
Image: ai_beautiful_girls_great_3698504292231033829_2.WEBP



Example 8: True: AI, Predicted: AI
Image: ai_dreamdesigns_3696922365447891659_5.JPG



Example 9: True: AI, Predicted: AI

Image: aindroidparanoidsfw_3697756815887945837_0.JPG



Example 10: True: AI, Predicted: AI
Image: chaipeau_3601980915332618198_10.jpeg



Example 11: True: AI, Predicted: Human
Image: cosmopolitan_hk_3696821183761525754_1 2.jpg





Example 12: True: AI, Predicted: AI
Image: fernlichtsicht_3748881358836495779_8.jpeg



Example 13: True: AI, Predicted: AI
Image: iam__pex_3685330416235124870_1.JPG



Example 14: True: AI, Predicted: AI
Image: junto.ai_3709410945325550214_1.jpeg





Example 15: True: AI, Predicted: AI
Image: kurotorauniverse_3685667117338655024_3.jpg

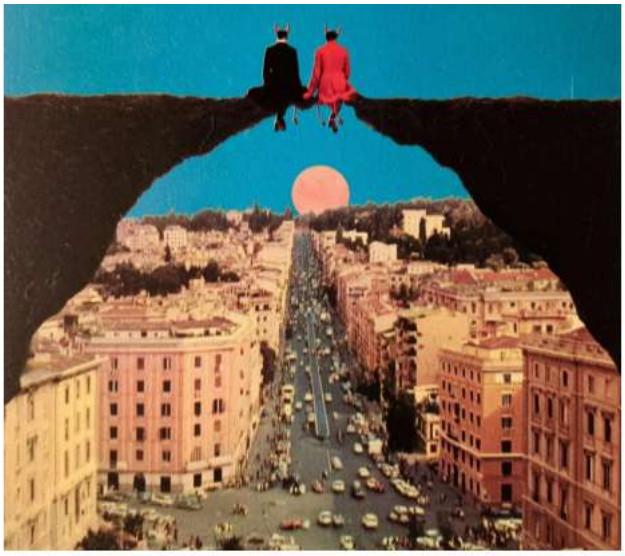


Example 16: True: AI, Predicted: AI
Image: manirostom_3207564369397354925_6.jpg



Example 17: True: AI, Predicted: AI
Image: neptunianglitterball_3697935375847511718_10.jpeg





Example 18: True: AI, Predicted: AI
Image: scinnart_3707783855162060358_2.jpg



Example 19: True: AI, Predicted: AI
Image: the.code.doodles_3706161526342990548_2.jpeg



Example 20: True: AI, Predicted: AI
Image: vayro.etc_3707060175661716355_5.WEBP



Example 21: True: Human, Predicted: AI
Image: 9honey_3408881526638704837.jpg



Example 22: True: Human, Predicted: AI
Image: _melaniatrumpisaqueen__3423990597700430086.jpg





Example 23: True: Human, Predicted: AI
Image: _persiancelebnews__3407097744241251916.jpg



Example 24: True: Human, Predicted: Human
Image: accesshollywood_3405703054573058003.jpg



Example 25: True: Human, Predicted: Human
Image: backgrid_usa_3407924396758986141.jpg





Example 26: True: Human, Predicted: Human
Image: behindtylerschair_340576318094344759 8.jpg



Example 27: True: Human, Predicted: Human
Image: behindtylerschair_340576318107763863 3.jpg

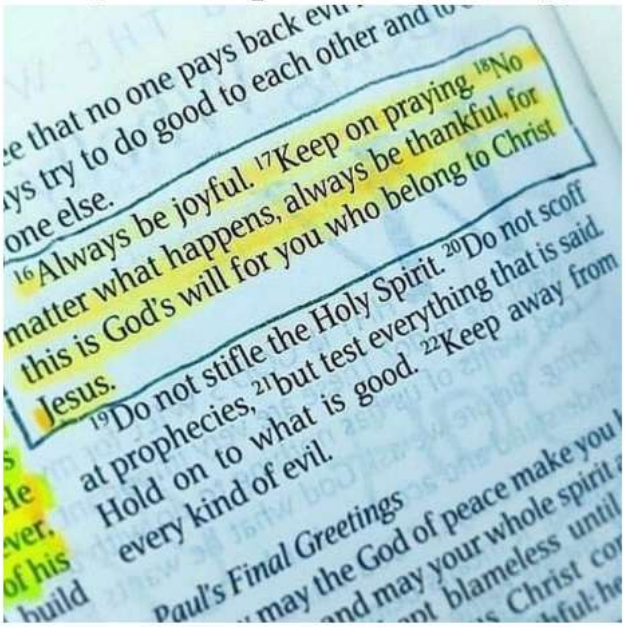


Example 28: True: Human, Predicted: AI
Image: bestpixmn_3343242988757186474.jpg





Example 29: True: Human, Predicted: AI
Image: bibleverse_3401966923598211997.jpg



Example 30: True: Human, Predicted: AI
Image: bitchy.tweets_3409134563956667366.jpg



Example 31: True: Human, Predicted: AI
Image: blast_3413067826958822932.jpg



Example 32: True: Human, Predicted: AI
Image: brobible_3408553476333558633.jpg



Example 33: True: Human, Predicted: AI
Image: bucketspot_3407315541615999929.jpg





Example 34: True: Human, Predicted: AI
Image: businessbounds_3407719596756074453.jpg

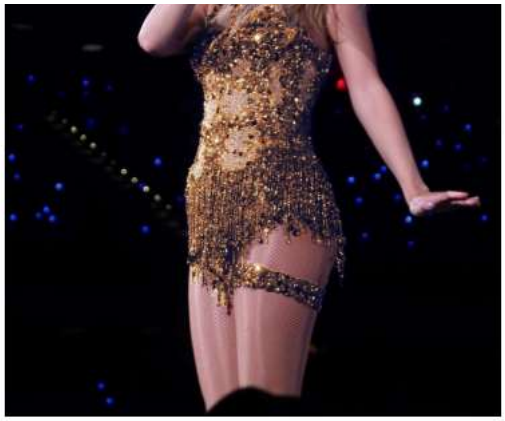


Example 35: True: Human, Predicted: AI
Image: bustle_3407854258367602039.jpg



Example 36: True: Human, Predicted: Human
Image: butterflyswift__3405610233156706977.jpg





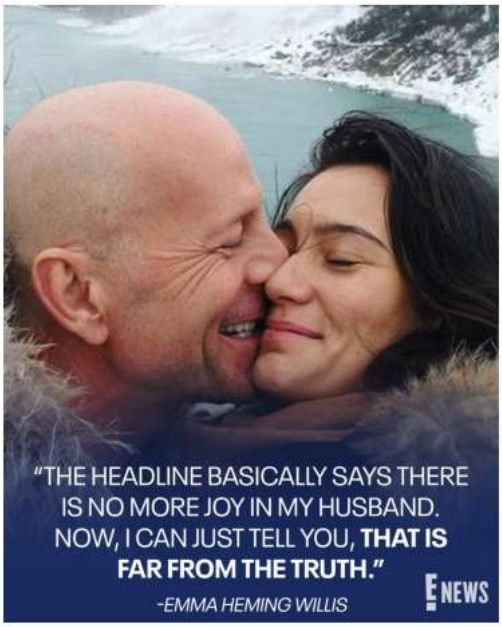
Example 37: True: Human, Predicted: AI
Image: carbonfinance_3347535747416435632.jpg



Example 38: True: Human, Predicted: Human
Image: elitedaily_3413154718173844386.jpg



Example 39: True: Human, Predicted: AI
Image: enews_3316012026165676652.jpg



Example 40: True: Human, Predicted: AI
Image: enews_3408648597939245435.jpg



Example 41: True: Human, Predicted: Human
Image: enews_3411410235343887914.jpg





Example 42: True: Human, Predicted: AI
Image: espn_3408565161907852033.jpg



Example 43: True: Human, Predicted: AI
Image: fanpagemarilynmonroeita_3404191709509419065.jpg



Example 44: True: Human, Predicted: AI
Image: fanpagemarilynmonroeita_3407127396906779341.jpg





Example 45: True: Human, Predicted: AI
Image: fashion_royal.ir_3408184405225864491.jpg



Example 46: True: Human, Predicted: AI
Image: geographybygeoff_3383249593838445804.jpg

